

LAB REPORT 02

BASIC SYNTAX, INPUT/OUTPUT, AND OPERATORS

1. OBJECTIVE

- To set up a basic program structure and manage user input and output.
- To apply correct data types based on the requirements.
- To follow indentation standards and use comments.
- To perform calculations using fundamental arithmetic operators.

2. THEORETICAL BACKGROUND

Basic Syntax and I/O. Every C++ program follows a specific boilerplate structure using the iostream library. The 'cout' object is used to print data to the screen using the stream insertion operator (<<). The 'cin' object reads data from the keyboard using the extraction operator (>>).

Data Types. C++ is a statically-typed language, meaning variables must be declared with a specific type before use. Common types include int (whole numbers), double (floating-point decimals), char (single characters), bool (true/false), and string (text).

Arithmetic Operators. C++ utilizes standard mathematical symbols for operations: + (Addition), - (Subtraction), * (Multiplication), / (Division), and % (Modulus, which returns the remainder of integer division).

3. SOFTWARE USED

- IDE: DEV-C++ IDE / VS Code
- Compiler: C++ Compiler

4. EXPERIMENTAL TASKS

Task 1: Variable Declaration and Memory Size

Declare different types of variables (int, double, char) and display their assigned values along with the memory size they occupy using the sizeof() operator.

```
#include <iostream>
using namespace std;

int main() {
    int age = 22;
    double salary = 45000.50;
    char grade = 'A';

    cout << "Value of age: " << age << " | Size: " << sizeof(age) << "
bytes" << endl;
    cout << "Value of salary: " << salary << " | Size: " << sizeof(salary)
<< " bytes" << endl;
    cout << "Value of grade: " << grade << " | Size: " << sizeof(grade) <<
" bytes" << endl;

    return 0;
}
```

Task 2: Floating-Point Declaration

Write a C++ program that declares and displays a floating-point variable called 'pi' and assigns it the value 3.14.

```
#include <iostream>
using namespace std;

int main() {
    // Declare a floating-point variable
    double pi = 3.14;

    // Display the variable
    cout << "The value of pi is: " << pi << endl;

    return 0;
}
```

Task 3: Temperature Conversion

Write a C++ program that prompts the user for a temperature in Celsius, converts it to Fahrenheit using the formula $F = (C * 9/5) + 32$, and displays the result.

```
#include <iostream>
using namespace std;

int main() {
    double celsius, fahrenheit;

    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    // Perform conversion
    fahrenheit = (celsius * 9.0 / 5.0) + 32.0;

    cout << celsius << " Celsius is equal to " << fahrenheit << "
    Fahrenheit." << endl;

    return 0;
}
```

5. CONCLUSION

This lab covered the core syntactical elements of C++. The implementation of basic input and output streams using `cin` and `cout` allowed for interactive programming. By declaring different data types and performing arithmetic operations, including temperature conversions and checking memory sizes, students gained a practical understanding of how data is stored, manipulated, and presented in C++.